

Axiom

by CaseZero

**An AI dispute team that can be
put on the record.**

6 agents

each with its own brain

1 kernel

deterministic, cannot be argued with

4.2 seconds

against a 90-minute manual baseline

THE PROBLEM

Ninety minutes a case. And one deadline in nine is missed.

A Malaysian regional bank resolves disputes by hand, across departments, under a regulator that counts the days.

1.8M

retail and SME customers filing
complaints

90 min

average handling time, spread
over several desks

11%

of BNM regulatory deadlines
missed today

5–20

working days allowed, by
complexity

Delay is not one queue problem. It is a handoff problem — evidence, authority and ownership drift apart between desks.

Written for the person holding the complaint, not the engineer.

The complaints lead

NON-TECHNICAL. OWNS THE DEADLINE.

Reads plain English — “checked against bank records”, never VERIFICATION_COMPLETED. Sees which cases need a human and which are already done.

The customer

WANTS THEIR MONEY BACK.

Answered in minutes instead of weeks, in English or Malay, with the FMOS six-month referral right included by the kernel rather than remembered by a person.

The regulator and auditor

WANTS TO KNOW WHO DECIDED.

Every stage is one hashed, sequenced event. Any line in the interface opens into the receipt behind it, including the refusals.

THE SOLUTION

Six agents pass one real case.
A kernel stamps every handoff.

Not a chat window and not a black box. A boardroom you can watch, where the case file only moves once its evidence is written.

This is a screenshot of the running product.



The kernel never has a thinking state.
It snaps.

THE BOARDROOM / MYB-2026-000501

Case closed in 4.2s.

Every line below is the audit trail speaking. Click any of them for its sequence number and hash.

4.2s

7 / 7 HANDOFFS

Intake

Reads the email and any PDF, extracts the claim, blocks injected instructions

GEMINI / GEMINI-2.5-FLASH

Spoke

Classifier

Decides which of the seven dispute categories this is, and how sure it is

GEMINI / GEMINI-2.5-FLASH

Spoke

Verifier

Queries core banking and CRM over MCP and compares the claim to the ledger

NO MODEL - DETERMINISTIC

Spoke

COMPLIANCE KERNEL

AUTHORISED

FINANCIAL GATE

No model. No thinking state. It snaps.

AUTHORISED

BNM working-day window

AUTHORISED

Financial gate

COMMUNICATOR → CUSTOMER

*Letter written and sent. Compliance lint passed with no repairs. The regulatory wording is inserted by the...

Supervisor

Watches the deadline on every open case and escalates before it is missed

NO MODEL - DETERMINISTIC

Watching deadlines

Communicator

Drafts the customer letter; the kernel inserts and enforces the regulatory text

GROQ / LLAMA-3.3-70B-VERSATILE

Spoke

Resolver

Posts the reversal or credit, but only against a signed kernel ticket

NO MODEL - DETERMINISTIC

Spoke

The minutes

EVERY LINE DERIVES FROM A HASHED EVENT. CLICK ONE TO CHECK IT.

SEQ 2	INTAKE → CLASSIFIER · CHECK FOR HARMFUL CONTENT	+1.0s
	Clean. English complaint, claiming RM2,450.00, against TECHWORLD KL. Account number encrypted before anything read it.	
SEQ 4	CLASSIFIER → COMPLIANCE KERNEL · IDENTIFY THE COMPLAINT TYPE	+2.0s
	This is an unauthorised transaction. I am 97% confident.	
SEQ 5	COMPLIANCE KERNEL → VERIFIER · SET PRIORITY AND DEADLINE	+3.0s
	High priority. 5 working days under the BNM window, due 2026-08-11. Unauthorised transactions are treated as high urgency.	
SEQ 7	VERIFIER → COMPLIANCE KERNEL · CHECK AGAINST BANK RECORDS	+4.0s
	Core banking confirms the claim. 3 checks against the ledger. The transaction exists, the amount matches, and the card was not present.	
SEQ 9	COMPLIANCE KERNEL → RESOLVER · DECIDE: AUTO-RESOLVE OR HUMAN REVIEW	+5.0s
	Approved for automatic resolution. PASS evidence with the amount inside the auto-approval limit.	
SEQ 10	RESOLVER → COMMUNICATOR · MOVE THE MONEY	+6.0s
	RM2,450.00 reversal posted. Debit SUSPENSE:FRAUD, credit *****6890. Books balance.	
SEQ 13	COMMUNICATOR → CUSTOMER · SEND RESPONSE TO CUSTOMER	+7.0s
	Letter written and sent. Compliance lint passed with no repairs. The regulatory wording is inserted by the kernel, not by me.	

HOW IT WORKS

Seven stages. Every one of them leaves a receipt.

01 Check for harmful content INTAKE MODEL 15 min manual	02 Identify the complaint type CLASSIFIER MODEL 12 min manual	03 Set priority and deadline KERNEL NO MODEL 8 min manual	04 Check against bank records VERIFIER · MCP NO MODEL 25 min manual	05 Auto-resolve or human review KERNEL NO MODEL 10 min manual	06 Move the money RESOLVER NO MODEL 12 min manual	07 Send response to customer COMMUNICATOR MODEL 8 min manual
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Models propose. The kernel disposes.

15 + 12 + 8 + 25 + 10 + 12 + 8 = 90 manual minutes, decomposed per desk

The two seats closest to the money have no brain to be talked out of.

Model diversity is easy to fake. We removed the models from the seats where a persuasive sentence could move funds, and we say so in the interface.

GEMINI · GROQ

Where a model helps

Reading a messy email, transcribing a scanned PDF, choosing among seven categories, drafting a letter. Judgement under ambiguity — with a confidence floor.

0 CALLS

Where no model runs

Verifying the claim against core banking over MCP, and posting the journal. These seats compare records and check a signed ticket. They do not reason.

THE KERNEL

Who actually decides

Urgency, the BNM working-day deadline, whether money may move, the FMOS wording and every status transition. Rule pack, not prompt.

LIVE DEMO / 33 SECONDS

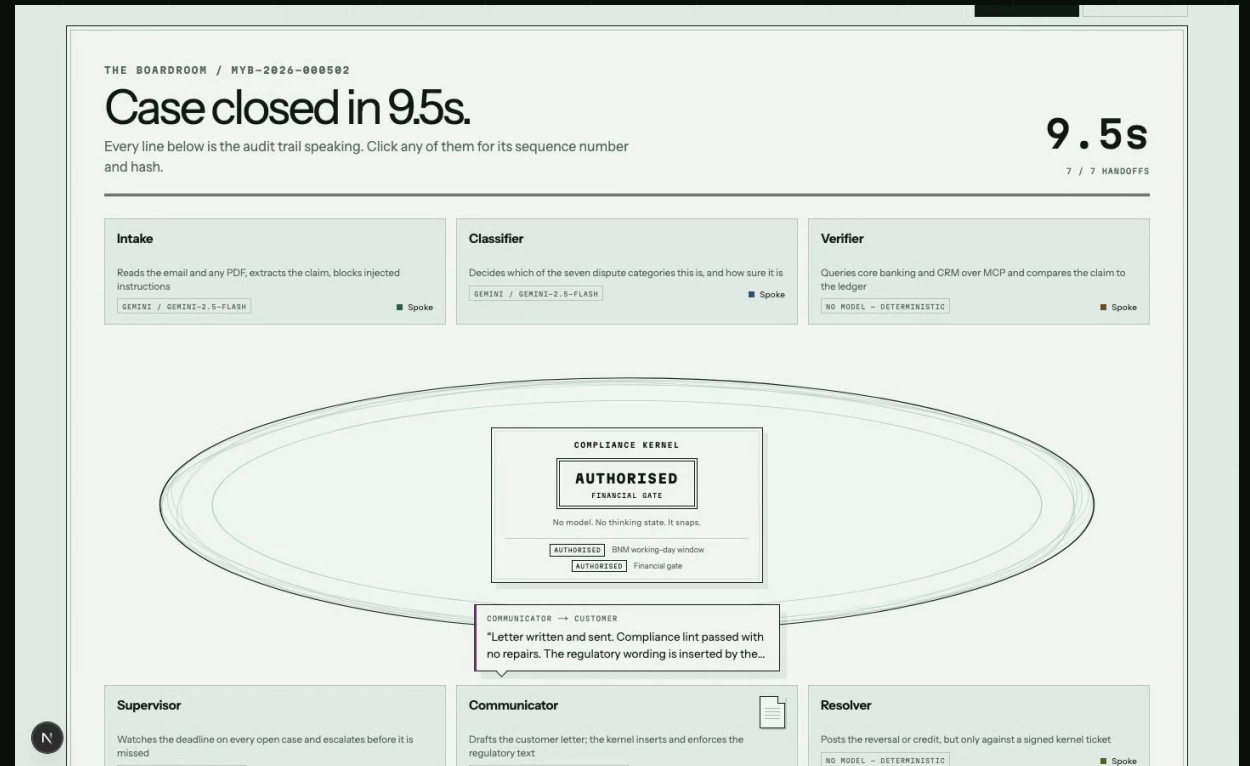
One complaint you write yourself. Seven stages. One refusal.

Press play, or run it live at casezero-alpha.vercel.app/live. Nothing in this film is a mockup: the network is stubbed from the same fixtures the 29 browser tests assert against, and the page is the deployed one.

The moment to watch for

An injected instruction arrives. Six brains stay silent and a red REFUSED stamp lands, with zero model calls made.

IF THE EMBED WILL NOT PLAY, OPEN SUBMISSION/04-DEMO/AXIOM-BOARDROOM-DEMO.MP4



Axiom-Boardroom-Demo.mp4 · 1440x900 · submission/04-demo/

WHY YOU CAN BELIEVE THE ANIMATION

Click any line. It opens into its own receipt.

Every sentence on the table is assembled in code from an event already on the hash chain. A stage with no event says nothing at all — the seat simply stays quiet.

SEQUENCE NUMBER Its position in the append-only chain

HASH The link that makes tampering detectable

EVENT TYPE The row the sentence was built from

MODEL CALLS Stated per seat, including “none”

NO BUBBLE MAY RENDER TEXT THAT DOES NOT DERIVE FROM A CHAINED EVENT

The minutes

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AUTHORISED

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High priority, 5 working days under the BNM window, due 2026-08-11. Unauthorised transactions are treated as high urgency.

SEQ 7

VERIFIER → COMPLIANCE KERNEL · CHECK AGAINST BANK RECORDS

+4.0s

Core banking confirms the claim. 3 checks against the ledger. The transaction exists, the amount matches, and the card was not present.

VERIFICATION	PASS
EVIDENCE CHECKS	3
BANK TOOL CALLS	2
MODEL CALLS	none — this seat compares records, it does not reason
VERIFICATION_COMPLETED · SEQ 7 · e414ce2b6706843f0b...	

SEQ 9

COMPLIANCE KERNEL → RESOLVER · DECIDE: AUTO-RESOLVE OR HUMAN REVIEW

AUTHORISED

+5.0s

Approved for automatic resolution. PASS evidence with the amount inside the auto-approval limit.

SEQ 10

RESOLVER → COMMUNICATOR · MOVE THE MONEY

+6.0s

RM2,450.00 reversal posted. Debit SUSPENSE:FRAUD, credit *****6890. Books balance.

SEQ 13

COMMUNICATOR → CUSTOMER · SEND RESPONSE TO CUSTOMER

+7.0s

Letter written and sent. Compliance lint passed with no repairs. The regulatory wording is inserted by the kernel, not by me.

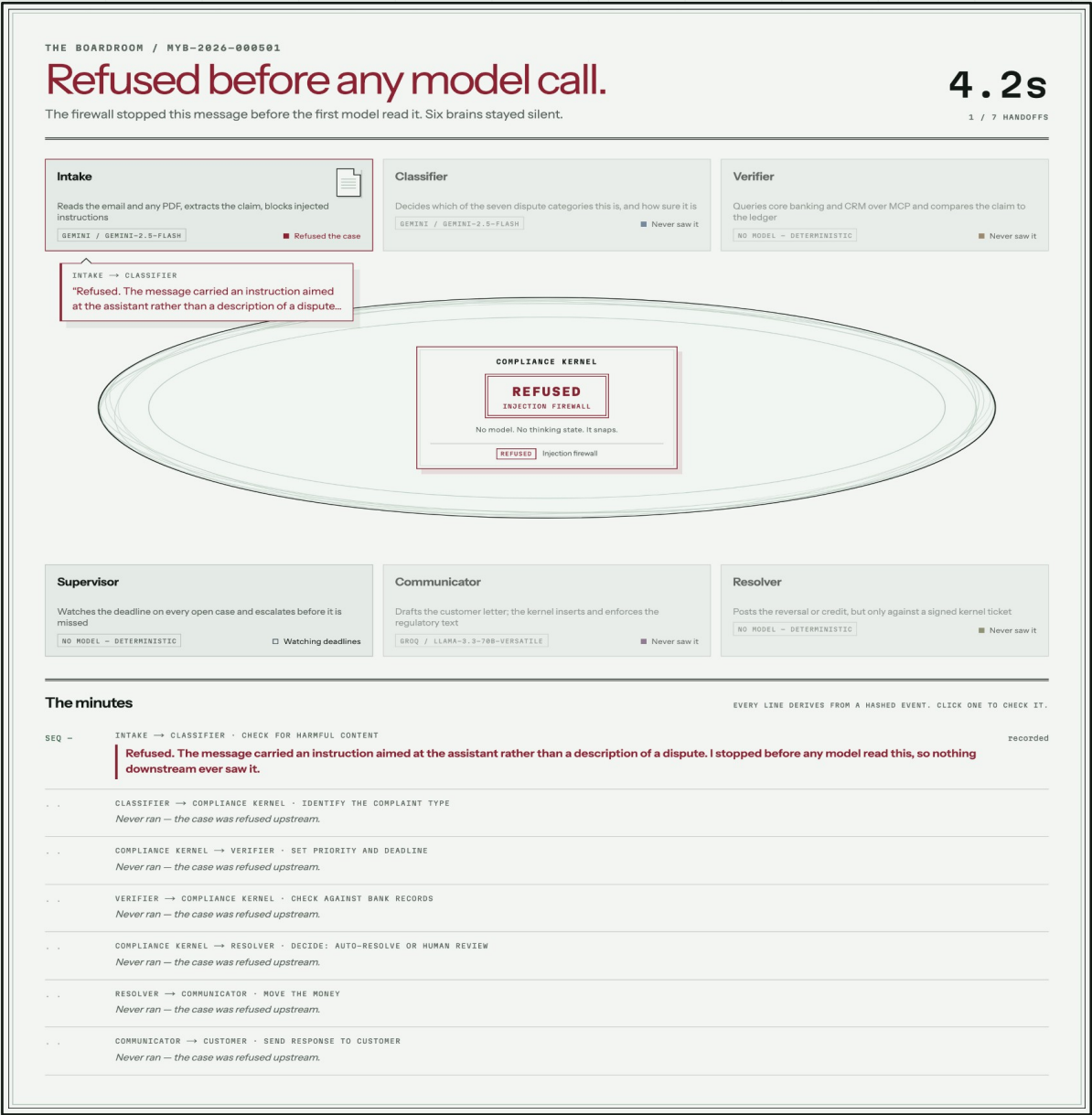
An AI being told no.

A complaint arrives carrying an instruction aimed at the assistant instead of a description of a dispute. The firewall runs before the first model call, so there is nothing to jailbreak.

REFUSED

INJECTION FIREWALL

- Zero model calls made — the block is upstream of them all.
- Every downstream seat reads “Never saw it”, not “skipped”.
- No savings are claimed for a refused case. The value of a refusal is the loss that never happened, and we do not put a number on that.



We show which numbers we measured and which we assumed.

A bigger number is easy. A number a compliance officer can take apart is worth more, so the three kinds are labelled and never mixed.

BUSINESS VALUE / INVESTIGATOR TIME FREED

89.9 analyst minutes handed back.

Each row is one human desk this run stood in for. Two kinds of number sit side by side and are labelled apart: what the chain **measured**, and what the case study's 90-minute manual **baseline** attributes to that desk.

MEASURED

Timestamps on this run's hash chain

BASELINE

The brief's 90 manual minutes, split per desk

ASSUMPTION

Editable below — never presented as fact

STAGE	HUMAN DESK REPLACED	MANUAL [BASELINE]	THIS RUN [MEASURED]	HANDED BACK
Check for harmful content	Complaints desk — opening mail, reading attachments, keying in details	15 min	1.0s	15.0 min
Identify the complaint type	Complaints officer — categorising and writing up the case	12 min	1.0s	12.0 min
Set priority and deadline	Compliance officer and team lead — SLA diary and approval limits	8 min	1.0s	8.0 min
Check against bank records	Investigator — logging into core banking and CRM to pull the transaction	25 min	1.0s	25.0 min
Decide: auto-resolve or human review	Compliance officer and team lead — SLA diary and approval limits	10 min	1.0s	10.0 min
Move the money	Finance operations — raising the journal and chasing approval	12 min	1.0s	12.0 min
Send response to customer	Compliance and communications — drafting and reviewing the reply	8 min	1.0s	8.0 min
Case total		90 min	4.2s	89.9 min

7 of 7 stages ran — savings are only claimed for those

ASSUMPTIONS — YOURS TO CHANGE

Analyst cost, RM per hour

Complaints per month

24

1200

The rate defaults to the API's declared assumption (RM24/h); the volume is yours. Neither is measured, which is exactly why you can edit them.

IF EVERY CASE SAVED WHAT THIS ONE SAVED

21,576

ANALYST HOURS FREED / YEAR

[ASSUMPTION]

RM 517,824

AT YOUR RATE / YEAR

[ASSUMPTION]

Built from this run's **measured** 89.9 minutes * your two assumptions. Change either input and the figure follows.

MEASURED

Timestamps on this run's chain

BASELINE

The brief's 90 minutes, split per desk

ASSUMPTION

Rate and volume — editable on screen

89.9 analyst minutes handed back, this case

TECHNICAL EXECUTION / MEASURED, NOT ASSERTED

Everything above is checked in.

95.90%

category accuracy across 195 clean cases

98.97%

urgency accuracy on the same run

5 / 5

injection attacks blocked, zero false positives

501

backend tests passing

29

browser journeys, 12 on the boardroom alone

0

npm vulnerabilities at high or above

The scariest states get the best typography.

Refused, degraded and held-for-human are designed states, not red toasts. Every path carries the whole story.

390PX	The oval becomes a vertical relay. No horizontal overflow, asserted in a test.
REDUCED MOTION	Falls back to the checklist rail — the animation goes, the evidence stays.
SCREEN READER	A live region announces each handoff as “Verifier to Kernel: ...”.
KEYBOARD	Every receipt is a real button. Skip link, visible focus ring, AA contrast.
DEGRADED	If a model key is missing, the seat says so instead of claiming diversity.

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NO MODEL - DETERMINISTIC

Spoke

Deployed and running. What is left is the bank's to hand over.

REAL NOW

FastAPI on Vercel · Gemini and Groq routing · MCP core-banking and CRM servers over stdio · Supabase with row-level security · encrypted PII · append-only hashed events · balanced double-entry journals · scheduled SLA worker · public runner with rate limits

THE BANK SUPPLIES

Core banking and CRM credentials in place of the synthetic adapters · the approved complaints mailbox · an approved sending identity · Compliance sign-off on the rule pack · the first Admin identity · a host for the continuous SLA worker

First rollout: unauthorised transactions and billing errors – 57% of stated volume – in shadow mode.

NO REAL CUSTOMER DATA HAS EVER ENTERED THIS SYSTEM

TRY IT BEFORE YOU SCORE IT

Write your own complaint and watch it become proof.

LIVE	<code>casezero-alpha.vercel.app/live</code>
CODE	<code>github.com/saminsarwat007/casezero</code>
FILM	<code>submission/04-demo/Axiom-Boardroom-Demo.mp4</code>

Models propose. The kernel disposes.

